MATEENAH JAHAN

Computational Biology · Biomedical Machine Learning · Bioinformatics

New Delhi, India · +91 95415 22732 · jahanmateenah55@gmail.com | Portfolio · GitHub · LinkedIn

PROFILE

Computational biology researcher working on reproducible machine learning for genomics and clinical decision support. **Fatima Fellow 2026, selected from 700+ applicants**; **ADA Consortium Discovery Prize winner**. My work spans immune-repertoire modeling, drug-response genomics, AI-driven drug repurposing, and clinical-LLM evaluation, and I build it to be reproducible — from raw data through to public, checkable tools.

Research interests: systems biology of cancer immunotherapy response — integrating immune-repertoire, perturbational, and multi-omic data to understand why responses last or fail, and building models whose predictions connect to real cellular behaviour and patient outcomes.

RESEARCH EXPERIENCE

Fatima Institute for Global AI Research — *Data Science Research Fellow (mentored by Dr. Marouen Ben Guebila, Harvard / Dana-Farber)* Apr 2026 – Present

Asking whether sex shapes how cancer cell lines respond to the same drugs, and whether any signal survives correction for lineage, dose, and tissue — work aimed at sex-aware precision dosing.

- Across the **GDSC drug-sensitivity panel (200+ drugs, 900+ cell lines)**, compared IC50 sensitivity between male- and female-derived lines with non-parametric testing and a **Random Forest** classifier to surface the largest sex-linked gaps. Repo →
- Cross-referenced **104k GTEx sex-biased gene–tissue associations (44 tissues)** against **596k CMAP perturbational drug experiments** using **Fisher's exact enrichment** with FDR correction, testing whether transcriptional sex bias and drug-response sex bias converge on the same biology. Live app →
- Built an interactive web app (live, on a public dataset) so the findings can be explored and challenged directly, rather than sitting as a static result.

CSIR–Institute of Genomics & Integrative Biology — *Research Trainee (CAR-T / TCR Repertoire) — supervised by Dr. Sherry Bhalla* Aug 2025 – Mar 2026

- Took a **DeepTCR** pipeline that no longer ran and rebuilt it into a working, reproducible workflow (**Python, TensorFlow 2.12**), getting all **11 tutorials** to run again. My dependency fixes were acknowledged in the official repository behind the 2021 **Nature Communications** paper. Tutorials →
- Trained deep-learning models predicting CD19 CAR-T response in refractory/relapsed B-cell lymphoma from **239,637 TCR sequences (34 patients)** with attention-based multiple-instance learning on H100 GPUs, reaching **AUC 0.754 ± 0.035 (100-fold Monte Carlo CV)**; surfaced responder-enriched CDR3 motifs and repertoire clusters. Paper →
- Wrote this up as the **first-author research report (CSIR-IGIB)**, covering methods, figures, and what the model could and could not show, and published the pipeline, a replication roadmap, and starter notebooks publicly. Deliverable →

PROFESSIONAL DATA EXPERIENCE

IUS Digital Solutions — *Data Analyst* Nov 2024 – Jul 2025

- Optimized MySQL/BigQuery reporting, **cutting report-generation time 40%** and improving data accuracy **25%**; built Python ETL over **10k+ records/month** and Tableau dashboards; customer-acquisition segmentation lifted conversion **15%**.

Google Ad Grants — NMI Program — *Data Consultant (Nonprofit Analytics)* Jun 2024 – Aug 2024

- Built GA conversion tracking and Apps Script reporting automation **saving 10+ hours/month**; statistical campaign models raised conversion **45%** for the supported non-profit.

Community Data & Teaching (BETTER HOPE NGO; independent) — *Education & Analytics in low-connectivity settings — Kashmir* 2022 – 2023

- Built an **offline-first assessment-tracking system for 100+ students** during extended internet shutdowns and a quantitative evaluation framework that **improved literacy scores 26%** over six months; designed low-bandwidth analytics for 10 local businesses with minimal digital infrastructure.

GRADUATE COURSEWORK — M.S. COMPUTER SCIENCE: ARTIFICIAL INTELLIGENCE & MACHINE LEARNING, WOOLF / SCALER NEOVARSITY

GPA 4.0/4.0, 100% average across 60 ECTS (12 graded courses). Coursework: High-Dimensional Data Analysis, Applied Statistics, Statistical Programming, Numerical Programming in Python, Foundations of Machine Learning, Introduction to Machine Learning, Advanced Machine Learning, Relational Databases (SQL), Data Visualisation Tools, Product Analytics, and Computer Programming I & II.

TECHNICAL SKILLS

Bioinformatics & Genomics — drug-response and perturbational genomics: GDSC drug-sensitivity analysis, CMAP/LINCS perturbational expression, GTEx sex-biased expression, and Fisher's exact enrichment.

Immune-repertoire and sequence modeling: TCR-repertoire modeling with DeepTCR, plus CDR3 motif and attention analysis.

Functional genomics and translation: GWAS-grounded drug repurposing, gene–drug interaction mapping, and clinical-LLM evaluation (Johns Hopkins Genomic Technologies & Python for Genomic Data Science).

Machine Learning & Stats: deep learning, attention-based multiple-instance learning, Monte Carlo cross-validation, Mann–Whitney & enrichment testing, imbalanced classification, dimensionality reduction, regularization, leakage-aware evaluation, model interpretation; A/B & hypothesis testing (ANOVA, chi-square).

Time Series & Recommenders: ARIMA / SARIMAX / Prophet, stationarity (ADF), seasonal decomposition; collaborative filtering, matrix factorization (SVD), KNN (RMSE / MAPE).

Programming & Tools: Python, SQL, C++, pandas, NumPy, statsmodels, scikit-learn, TensorFlow, PyTorch, HuggingFace Transformers, MySQL, BigQuery, Docker, Git, Tableau, Google Data Studio, FastMCP / Model Context Protocol.

SELECTED PUBLIC OUTPUTS

- **Biomni-AD — AI drug repurposing (ADA Consortium Discovery Prize, Round 1 winner):** deployed Biomni agents on the ADDI AD Workbench, cross-referencing **21 AD GWAS risk genes** against drug–target databases to rank repurposing candidates with pathway-level validation. Repo →
- **MedGemma blind-spot dataset (Hugging Face, CC-BY-4.0):** curated **13 reproducible probes** exposing **5 structural failure modes** of Google's MedGemma-4B (PT + IT) in hematologic oncology / CAR-T reasoning, with the correct clinical references for each case. Dataset →
- **Synapse / BoldConnect — MCP server for ligand discovery:** a FastMCP server that lets AI agents run GPCR ligand screening and mutation-impact checks, packaged to run in Docker with an adapter for the BOLD-GPCRs model. Repo →

EDUCATION & HONORS

- **Indira Gandhi National Open University** — M.A. Political Science (2019), First Division, **IEE U.S. equivalency 3.50/4.0.** **University of Kashmir** — B.Sc. Chemistry, Botany & Zoology (2017), **IEE U.S. equivalency 3.77/4.0.**
- **Fatima Fellow, 2026** (700+ applicants) · **ADA Consortium Discovery Prize, Round 1 winner** (Biomni-AD) · Merit Scholarship, M.S. CS (AI & ML) · **IELTS Academic 7.0 / CEFR C1** (Jun 2026) · Languages: English, Urdu.